



• AI & PRODUCTIVITY • LESSON 02

Understanding ChatGPT

How It Works, What It Can Do & How to Use It

Lesson 2 of 30 | AI & Productivity Program | Beginner

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- What ChatGPT is and how large language models work
- GPT model history — from GPT-1 to GPT-4o explained
- How to write better prompts and get reliable outputs
- Real-world use cases: writing, coding, analysis, research
- What ChatGPT gets wrong — hallucinations and limitations

01 | What Is ChatGPT?

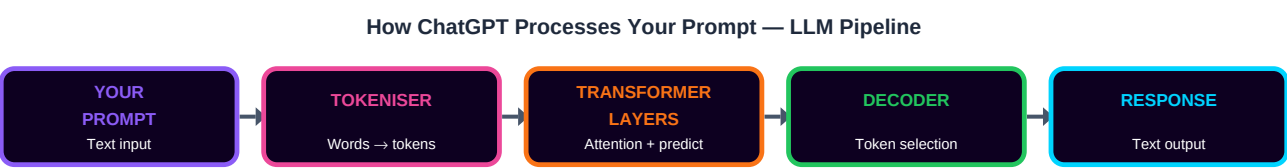
ChatGPT is a **conversational AI assistant** built by OpenAI, first released to the public in November 2022. It is powered by a **Large Language Model (LLM)** — a deep learning system trained on hundreds of billions of words of text from the internet, books, code, and other sources. ChatGPT can hold multi-turn conversations, write text, generate and explain code, translate languages, summarise documents, and reason through complex problems.

In its first week, ChatGPT gained one million users — faster than any product in history. It reached **100 million users in two months**, a record that stood until competitors began launching their own models. Today, ChatGPT is used by over 200 million people weekly for work, study, and creative tasks.

Milestone	Date	What Happened
GPT-1	June 2018	117M parameters. Proved language models could be pre-trained then fine-tuned for tasks.
GPT-2	Feb 2019	1.5B parameters. OpenAI initially withheld it, calling it "too dangerous to release."
GPT-3	May 2020	175B parameters. First model to achieve human-level performance on many NLP benchmarks. API access only.
InstructGPT / RLHF	Jan 2022	Introduced RLHF (Reinforcement Learning from Human Feedback) — made models follow instructions reliably.
ChatGPT (GPT-3.5)	Nov 2022	Public chatbot interface. 1M users in 5 days. Triggered the global AI boom.
GPT-4	Mar 2023	Multimodal (text + images). Passed bar exam in top 10%. Integrated into Bing, GitHub Copilot.
GPT-4o ("Omni")	May 2024	Combined text, voice, and vision in one model. Real-time voice conversation. 2× faster than GPT-4.
GPT-4o mini	Jul 2024	Small, fast, cheap model. Replaced GPT-3.5 Turbo as the default affordable option.

02 | How ChatGPT Works — Inside the LLM

ChatGPT is not a database that looks up answers. It is a **statistical prediction machine** — it predicts the most likely next word (token) given everything before it. This happens through a **transformer neural network** with billions of learned parameters that were adjusted during training to reflect patterns in human language.



KEY INSIGHT — ChatGPT Predicts, It Does Not Think

When ChatGPT writes "The capital of France is Paris," it is not retrieving a fact — it is predicting that "Paris" is the statistically most likely token to follow "The capital of France is." This is why ChatGPT can be confidently wrong — a phenomenon called hallucination. It generates plausible-sounding text, not verified truth. Always verify factual claims independently.

03 | GPT Model Versions — Which One Should You Use?

OpenAI releases multiple model versions simultaneously. Each differs in reasoning ability, speed, cost, and context window (how much text it can process at once). Choosing the right model for your task saves money and gets better results.

GPT Model Family — Key Versions and Capabilities

GPT-3.5
Turbo

Released: 2022
175B params
Fast & cheap
Good for simple tasks

GPT-4

Released: 2023
Multimodal
Smarter reasoning
Code + image input

GPT-4o

Released: 2024
Omni model
Text + voice + image
Fastest GPT-4 tier

GPT-4o mini

Released: 2024
Small & fast
Cost-efficient
API & daily use

Model	Context Window	Best For	Access	Relative Cost
GPT-3.5 Turbo	16K tokens	Simple tasks, high volume, chatbots, quick summaries	Free tier / API	Very low
GPT-4	128K tokens	Complex reasoning, nuanced writing, difficult code, analysis	ChatGPT Plus (\$20/mo) / API	High
GPT-4o	128K tokens	Multimodal tasks (images + text), voice, fastest GPT-4 quality	ChatGPT Plus / API	Medium-High
GPT-4o mini	128K tokens	Fast everyday tasks, API-driven apps, cost-sensitive projects	Free tier (limited) / API	Low
o1 / o3 (reasoning)	200K tokens	Math, science, logic problems requiring deep step-by-step reasoning	ChatGPT Pro / API	Very High

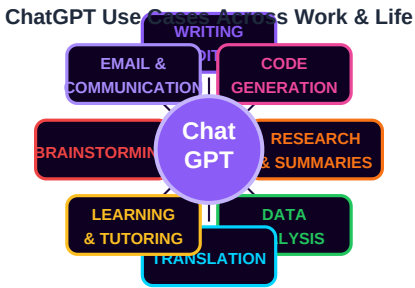
04 | Effective Prompting — How to Get Better Outputs

The quality of ChatGPT's output depends almost entirely on the quality of your prompt. A vague prompt gets a vague answer. A structured, specific prompt with context and constraints gets precise, useful output. These techniques apply to every AI model, not just ChatGPT.

Technique	What to Do	Example
Be specific	State exactly what you want, the format, and the length	"Write a 3-bullet summary of this text for a non-technical audience" — not "summarise this"
Give context	Tell the model who you are, what you're doing, and why	"I am a civil engineering student preparing for an exam. Explain prestressed concrete simply."
Assign a role	Tell ChatGPT to act as an expert persona	"Act as an experienced Python developer. Review this code for bugs and suggest improvements."
Use examples (few-shot)	Show the model the format you want with 1–3 examples	"Classify these reviews as Positive/Negative. Example: 'Great product!' → Positive."

Chain of thought	Ask the model to reason step by step	"Solve this problem step by step, showing each calculation." — improves math accuracy.
Constrain the output	Specify format: JSON, table, bullet list, word count, language	"Return the answer as a JSON object with keys: name, role, salary."
Iterate and refine	Treat the first response as a draft — follow up to improve it	"Good, now make it more concise." / "Rewrite this for a Year 10 student."

05 | Real-World Use Cases — ChatGPT Across Professions



Profession / Task	How ChatGPT Helps	Time Saved
Engineers	Explain technical concepts, write technical reports, generate code for calculations, draft RFIs and specifications	2–4 hrs/day on documentation
Students	Explain difficult concepts in simple terms, create study guides, solve example problems step-by-step, essay feedback	1–3 hrs/day on comprehension
Content Creators	Draft scripts, generate video titles and descriptions, write social media captions, repurpose content across formats	3–5 hrs/week on drafts
Developers / Coders	Generate boilerplate code, explain error messages, suggest fixes, write unit tests, document functions	30–60% reduction in debugging time
Business Professionals	Draft emails, create meeting agendas, summarise long documents, prepare presentations, market research	1–2 hrs/day on communication
Researchers	Summarise academic papers, identify key arguments, suggest related literature, explain statistical methods	50–70% reduction in literature review time

06 | Limitations — What ChatGPT Cannot Do Reliably

Limitation	Why It Happens	How to Mitigate
Hallucinations	Predicts plausible tokens — not verified facts. Will confidently invent citations, statistics, and events.	Never use ChatGPT as sole source for facts. Verify all specific claims independently.
Knowledge cutoff	Trained on data up to a fixed date. Has no knowledge of recent events unless given search tools.	Use ChatGPT with web browsing enabled, or provide the latest information in your prompt.
No real-time data	Cannot access live prices, news, stock data, or current information without plugins/tools.	Paste relevant current data directly into the prompt for analysis.
Maths errors	Standard GPT models predict numbers like words — can make arithmetic and algebra errors.	Use o1/o3 reasoning models for maths. Use Code Interpreter to run actual calculations.

No persistent memory (default)	Starts fresh each session — no memory of past conversations unless memory is enabled.	Enable ChatGPT memory, or paste relevant context at the start of each session.
Sycophancy	Tends to agree with the user even when wrong — will validate incorrect premises.	Ask "What's wrong with my approach?" explicitly. Use adversarial prompts.
Context window limits	Very long documents may exceed context limit — early text gets "forgotten."	Use chunking: process large documents in sections; use models with 128K+ context windows.

Reflection:

"Open ChatGPT and test it with two prompts on the same topic: one vague ("Tell me about solar energy") and one structured ("Act as a solar engineer. Explain in 5 bullet points the key factors a first-time installer must consider when sizing a residential off-grid solar system in a tropical climate."). Compare the quality and usefulness of the two responses."

Action checklist:

Next up — AI-003:

Prompt Engineering Basics — the frameworks, techniques, and [t.me/ayetechub](#) patterns that consistently produce great AI outputs.

Community & support: